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Assessment: Assessment Task No. 1 - Introduction to Machine Learning

From Data to Prediction: Understanding Machine Learning-Based Social Media Recommendation

Chosen ML Application:

Social Media Recommendation - a system that suggests people or accounts that a user may want to follow based on their interests, activities, connections, and previous interactions.

Problem Description:

Social media applications have numerous numbers of users and consequently numerous accounts on these applications. It becomes very difficult for users to navigate through such a long list of potential accounts to follow and end up with those they are most likely to be interested in. The social media recommendation system can help by analyzing a user's activity, interests, connections and interactions to come up with potential accounts that a user might be interested to follow based on these activities. The recommendation system would therefore ease the process of exploring potential accounts on a social media application.

Input and Output Identification:

The table below identifies the application, problem, inputs, outputs, patterns, and data needed.

Question	Your Answer
1. What is the application?	Social media recommendation. For example, a system that suggests people or accounts that a user may want to follow.
2. What problem are you trying to solve?	Recommend relevant people or accounts to a user based on their interests, connections, and previous activities.
3. What are the inputs?	Number of mutual friends, shared interests, followed pages or groups, previous interactions, and user activities.
4. What is the expected output?	A list of recommended people or accounts that the user may be interested in following.
5. What patterns or relationships might the ML system learn?	Users with more mutual friends and shared interests may be more likely to follow or connect with each other.

6. What data would the system need?	User profiles, friendship/following information, mutual friend data, user interests, groups and pages followed, previous interactions, and historical follow/connection data.

Traditional Programming vs. Machine Learning:

In traditional programming, the program is written in a way that gives instructions of what the computer should do. The flow of traditional programming can be depicted as:

Input → Algorithm/rules → Output

For a social media recommendation, a programmer would have to write rules like: **"if the mutual friends between user 1 and 2 are more than 5, recommend this account" or "if the common interests are equal to or more than 3, recommend this account"**. There are too many rules that a programmer would have to write down for different cases.

For example, if a user has **6 mutual friends and 4 shared interests**, the programmed rules could recommend that account.

In Machine Learning, the flow is different. It uses Input + Output/Data → Machine Learning Algorithm → Learned Relationship. Instead of manually creating every rule, the algorithm is given many examples of previous user behavior. It studies the inputs and outputs and learns the relationship between them. After training, the model can use new information to make a prediction.

Machine learning is better in this case since social media recommendations are not easily predictable.

Mini Dataset:

Below is a simple dataset showing the relationship between the number of mutual friends, shared interests, and whether the user followed the recommended account.

Observation	Mutual Friends	Shared Interests	Followed?
1	8	5	Yes
2	6	4	Yes
3	2	1	No
4	3	2	No
5	7	4	Yes

6	1	1	No
7	5	3	Yes
8	4	1	No

Pattern/Relationship:

The pattern that I notice in the dataset is that users with more mutual friends and shared interests are generally more likely to follow the recommended account.

For example, users with 5 or more mutual friends and 3 or more shared interests mostly resulted in "Yes." On the other hand, users with fewer mutual friends and shared interests mostly resulted in "No."

For a new example:

Mutual Friends	Shared Interests	Followed?
7	5	**Yes**

The predicted output is Yes because the user has many mutual friends and shared interests, which is similar to the examples that resulted in "Yes."

For this simple dataset, we can represent the relationship using a basic scoring function:

Recommendation Score = Number ofMutual Friends + Number of Shared Interests

For example:

7 + 5 = 12. So the recommendation score is 12.

A higher value in this simplified example means that the user is more likely to follow the recommended account. This is just a toy model to show how inputs are related to the output in this mini-dataset. In reality, a social media recommendation algorithm would detect more nuanced patterns in the data in order to create a more accurate recommendation.

Explanation of the Training Process:

How can a Machine Learning algorithm learn from data?

First, the algorithm needs some data to learn from: the number of mutual friends and shared interests as input and information on whether the user followed the suggested account or not as an output. Then, the algorithm is trained on the data, learning the patterns that correlate certain input data with the output results. The algorithm thus learns to recognize the relationship between input data and output results, constantly adjusting the model until the predicted output matches the actual output. Finally, with a sufficiently trained algorithm, one can use it to predict the output from a different data sample, unseen by the algorithm.

The process is:
Input → Output → Training → Learning the Relationship → Prediction

For this example:
Mutual Friends + Shared Interests → Followed/Not Followed → Training → Learn User Behavior → Predict Follow/No Follow

Short Reflection:

Why is learning from data useful instead of manually programming every possible rule?

Why is learning from data useful instead of manually programming every possible rule? Learning from data is useful because real-world situations can be too complicated to describe using individual rules. In social media, users have different interests, behaviors, friends, and interactions, so creating a rule for every possible situation would be difficult and time-consuming. Machine Learning allows the computer to discover patterns from many examples and use those patterns to make predictions. Thus, learning from data is more practical for recommendation systems than manually programming every possible rule.